

Deep Learning: Homework 2

1. Your task is to find a linear approximation of the function $\sqrt{1+x}$, $x \in [0, 1]$. Your homework should contain the following steps:

- a) Generate $N = 10000$ random numbers from $[0, 1]$:

$$x_1, x_2, \dots, x_N \in [0, 1],$$

and then obtain their labels: $y_i = \sqrt{1+x_i}$, $i = 1, 2, \dots, N$.

- b) Do linear regression on your generated data using stochastic gradient descent (implement yourself).
- c) Do linear regression on your generated data using mini-batch gradient descent (implement yourself).
- d) Sketch the graphs of all approximations on one graph.
- e) Compare all solutions with the first degree Taylor approximation of the function $\sqrt{1+x}$.

- 2*. Pass the following [tutorial](#).